

Soumya suryawanshi (24070521062) Software Copy - Experiment 3

1. Experiment Title

Implement control structures and form validation; create a grading system based on user-entered marks.

2. Software/Tools Required

Visual Studio Code

HTML5

CSS3

JavaScript

Google Chrome

3. Experiment Program Code

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SIT Student Portal - Grading System</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <!-- HEADER -->
  <div class="header">
    <div class="header-container">

      <div class="logo-section">
        <div class="logo">SIT</div>

        <div class="brand-info">
          <h1>Symbiosis Institute of Technology</h1>
          <p>Symbiosis International (Deemed University), Pune</p>
        </div>
      </div>

      <nav class="nav-links">
        <a href="#">Home</a>
        <a href="#">Admissions</a>
        <a href="#">Academics</a>
        <a href="#">Contact</a>
      </nav>

    </div>
```

</div>

<!-- GRADING SECTION -->

<div class="grading-wrapper">

<div class="grading-container">

<div class="grading-header">

<h2>Student Marks Grading System</h2>

</div>

<!-- Input Form -->

<form id="gradingForm" class="grading-form">

<div class="form-row">

<input
type="text"
id="studentName"
placeholder="Student Name"
required>

<input
type="number"
id="math"
placeholder="Maths (0-100)"
min="0"
max="100"
required>

<input
type="number"
id="science"
placeholder="Science (0-100)"
min="0"
max="100"
required>

<input
type="number"
id="english"
placeholder="English (0-100)"
min="0"
max="100"
required>

```
<input
  type="number"
  id="history"
  placeholder="History (0-100)"
  min="0"
  max="100"
  required>
```

```
</div>
```

```
<button
  type="submit"
  class="btn-add">
```

Add Record

```
</button>
```

```
</form>
```

```
<!-- Results -->
```

```
<div class="results-area">
```

```
<h3>Student Results</h3>
```

```
<table class="grade-table">
```

```
<thead>
```

```
<tr>
```

```
<th>Name</th>
```

```
<th>Total</th>
```

```
<th>%</th>
```

```
<th>Grade</th>
```

```
<th>Status</th>
```

```
<th>Action</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody id="resultBody">
```

```
</tbody>
```

```
</table>
```

```
<div
```

```
id="emptyMsg"
class="empty-msg">
```

No records found. Add a student above.

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- FOOTER -->
```

```
<div class="footer">
```

```
<div class="footer-container">
```

```
<div class="footer-col">
```

```
<h4>Symbiosis Institute of Technology</h4>
```

```
<p>Senapati Bapat Road, Pune - 411016</p>
```

```
<p>Maharashtra, India</p>
```

```
</div>
```

```
<div class="footer-col">
```

```
<h4>Quick Links</h4>
```

```
<ul>
```

```
<li><a href="#">Notice Board</a></li>
```

```
<li><a href="#">Timetable</a></li>
```

```
<li><a href="#">Examination</a></li>
```

```
</ul>
```

```
</div>
```

```
<div class="footer-col">
```

```
<h4>Connect</h4>
```

```
<ul>
```

```

        <li><a href="#">Facebook</a></li>
        <li><a href="#">LinkedIn</a></li>
        <li><a href="#">Twitter</a></li>

    </ul>

</div>

</div>

<div class="footer-bottom">
    <p>&copy; 2026 Symbiosis Institute of Technology. All Rights Reserved.</p>
</div>

</div>

<script src="script.js"></script>

</body>
</html>

```

Script.js

```

const form = document.getElementById('gradingForm');
const resultBody = document.getElementById('resultBody');
const emptyMsg = document.getElementById('emptyMsg');

let students = JSON.parse(localStorage.getItem('students')) || [];

// Load records when page opens
window.onload = function () {
    renderTable();
};

// Add Student
form.addEventListener('submit', function (e) {
    e.preventDefault();

    const name = document.getElementById('studentName').value.trim();
    const math = parseFloat(document.getElementById('math').value) || 0;
    const science = parseFloat(document.getElementById('science').value) || 0;
    const english = parseFloat(document.getElementById('english').value) || 0;
    const history = parseFloat(document.getElementById('history').value) || 0;

    if (name === "") {
        alert("Please enter student name.");
        return;
    }

```

```
if (
  math > 100 || science > 100 ||
  english > 100 || history > 100
) {
  alert("Marks cannot exceed 100.");
  return;
}

const total = math + science + english + history;
const percentage = (total / 400) * 100;

const grade = calculateGrade(percentage);
const status = percentage >= 40 ? "Pass" : "Fail";

const student = {
  id: Date.now(),
  name,
  total,
  percentage: percentage.toFixed(1),
  grade,
  status
};

students.push(student);

localStorage.setItem(
  "students",
  JSON.stringify(students)
);

form.reset();

renderTable();
});

// Grade Calculation
function calculateGrade(percentage) {

  if (percentage >= 80) {
    return "A";
  }

  if (percentage >= 60) {
    return "B";
  }
}
```

```

    if (percentage >= 40) {
        return "C";
    }

    if (percentage >= 33) {
        return "D";
    }

    return "F";
}

// Display Table
function renderTable() {

    resultBody.innerHTML = "";

    if (students.length === 0) {
        emptyMsg.style.display = "block";
        return;
    }

    emptyMsg.style.display = "none";

    students.forEach(function (student, index) {

        const row = document.createElement("tr");

        row.innerHTML =
            `<td><strong>${student.name}</strong></td>
            <td>${student.total}</td>
            <td>${student.percentage}%</td>
            <td><span class="grade grade-${student.grade}">
                ${student.grade}
            </span></td>
            <td><span class="status status-${student.status}">
                ${student.status}
            </span></td>
            <td>
                <button
                    class="btn-del"
                    onclick="deleteStudent(${index})">
                        &times;
                </button>
            </td>`;

        resultBody.appendChild(row);
    });
}

```

```

});

}

// Delete Student
function deleteStudent(index) {

    if (confirm("Delete this record?")) {

        students.splice(index, 1);

        localStorage.setItem(
            "students",
            JSON.stringify(students)
        );

        renderTable();
    }
}

```

4. Output

The screenshot displays the Symbiosis Institute of Technology Student Portal. The main content area features a form titled "Enter Student Marks" with a "Logout" button. The form includes input fields for "Student Name", "Maths (0-100)", "Science (0-100)", "English (0-100)", and "History (0-100)", along with an "Add Record" button. Below the form is a table titled "Student Results" with columns: Name, Total, %, Grade, Status, and Action. The table contains one entry for "Soumya" with a total of 352, 88.0%, Grade A, Status Pass, and an Action button (red circle with a plus sign).

Name	Total	%	Grade	Status	Action
Soumya	352	88.0%	A	Pass	

The footer of the page includes the Symbiosis Institute of Technology logo and contact information, a "Quick Links" section with links to Notice Board, Timetable, and Examination, and a "Connect" section with links to Facebook, LinkedIn, and Twitter. The copyright notice at the bottom reads: "© 2025 Symbiosis Institute of Technology. All Rights Reserved."

5. Case Study Title

Student Portal Login Page with Password Validation System

6. Case Study Program Code

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>SIT Student Portal - Login</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <!-- HEADER -->
  <div class="header">
    <div class="header-container">

      <div class="logo-section">
        <div class="logo">SIT</div>

        <div class="brand-info">
          <h1>Symbiosis Institute of Technology</h1>
          <p>Symbiosis International (Deemed University), Pune</p>
        </div>
      </div>

      <nav class="nav-links">
        <a href="#">Home</a>
        <a href="#">Admissions</a>
        <a href="#">Academics</a>
        <a href="#">Contact</a>
      </nav>

    </div>
  </div>

  <!-- LOGIN SECTION -->

  <div id="loginSection" class="login-wrapper">

    <div class="login-card">

      <div class="card-top-border"></div>
```

```
<div class="card-content">
```

```
<h2>Student Portal Access</h2>
```

```
<p>
```

```
  Enter your Enrollment Number and Password  
  to continue.
```

```
</p>
```

```
<form id="loginForm">
```

```
<div class="form-group">
```

```
<label for="username">
```

```
  Enrollment Number
```

```
</label>
```

```
<input
```

```
  type="text"
```

```
  id="username"
```

```
  placeholder="24070521062"
```

```
  required>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="password">
```

```
  Password
```

```
</label>
```

```
<input
```

```
  type="password"
```

```
  id="password"
```

```
  placeholder="●●●●●●●●"
```

```
  required>
```

```
</div>
```

```
<div class="form-options">
```

```
<label class="remember-me">
```

```
<input type="checkbox">
```

```
  Remember me
```

</label>

Forgot Password?

</div>

<button
type="submit"
class="btn-login">

LOGIN

</button>

</form>

<div id="loginMessage" class="message"></div>

<div class="support-text">

<p>

For technical support contact

helpdesk@sitpune.edu.in

</p>

</div>

</div>

</div>

</div>

<!-- FOOTER -->

<div class="footer">

```
<div class="footer-container">
```

```
<div class="footer-col">
```

```
<h4>Symbiosis Institute of Technology</h4>
```

```
<p>Senapati Bapat Road, Pune - 411016</p>
```

```
<p>Maharashtra, India</p>
```

```
</div>
```

```
<div class="footer-col">
```

```
<h4>Quick Links</h4>
```

```
<ul>
```

```
<li><a href="#">Notice Board</a></li>
```

```
<li><a href="#">Timetable</a></li>
```

```
<li><a href="#">Examination</a></li>
```

```
</ul>
```

```
</div>
```

```
<div class="footer-col">
```

```
<h4>Connect</h4>
```

```
<ul>
```

```
<li><a href="#">Facebook</a></li>
```

```
<li><a href="#">LinkedIn</a></li>
```

```
<li><a href="#">Twitter</a></li>
```

```
</ul>
```

```
</div>
```

```
</div>
```

```
<div class="footer-bottom">
```

```
<p>
```

```
&copy; 2026 Symbiosis Institute of Technology.
```

All Rights Reserved.

</p>

</div>

</div>

<script src="script.js"></script>

</body>

</html>

Script.js

```
const form = document.getElementById('gradingForm');
const resultBody = document.getElementById('resultBody');
const emptyMsg = document.getElementById('emptyMsg');
```

```
let students = JSON.parse(localStorage.getItem('students')) || [];
```

```
// Load existing records
window.onload = function () {
  renderTable();
};
```

```
// Add Student
form.addEventListener('submit', function (e) {
```

```
  e.preventDefault();
```

```
  const name = document.getElementById('studentName').value.trim();
  const math = parseFloat(document.getElementById('math').value) || 0;
  const science = parseFloat(document.getElementById('science').value) || 0;
  const english = parseFloat(document.getElementById('english').value) || 0;
  const history = parseFloat(document.getElementById('history').value) || 0;
```

```
  if (name === "") {
    alert("Please enter student name.");
    return;
  }
```

```
  if (
    math > 100 ||
    science > 100 ||
    english > 100 ||
    history > 100
  ) {
    alert("Marks cannot exceed 100.");
```

```

    return;
}

const total = math + science + english + history;
const percentage = (total / 400) * 100;

const grade = calculateGrade(percentage);
const status = percentage >= 40 ? "Pass" : "Fail";

const student = {
  id: Date.now(),
  name,
  total,
  percentage: percentage.toFixed(1),
  grade,
  status
};

students.push(student);

localStorage.setItem(
  "students",
  JSON.stringify(students)
);

form.reset();

renderTable();
});

// Grade Calculation
function calculateGrade(percentage) {

  if (percentage >= 80) return "A";
  if (percentage >= 60) return "B";
  if (percentage >= 40) return "C";
  if (percentage >= 33) return "D";

  return "F";
}

// Display Table
function renderTable() {

  resultBody.innerHTML = "";

  if (students.length === 0) {

```

```

    emptyMsg.style.display = "block";
    return;
}

emptyMsg.style.display = "none";

students.forEach(function (student, index) {

    const row = document.createElement("tr");

    row.innerHTML =
        ` <strong>${student.name}</strong></td>         <td>${student.total}</td>         <td>${student.percentage}%</td>         <td><span class="grade grade-${student.grade}">${student.grade}</span></td>         <td><span class="status status-${student.status}">${student.status}</span></td>         <td>             <button class="btn-del"             onclick="deleteStudent(${index})">                 &times;             </button>         </td>`;      resultBody.appendChild(row);  });  }  // Delete Student function deleteStudent(index) {      if (confirm("Delete this record?")) {          students.splice(index, 1);          localStorage.setItem(             "students",             JSON.stringify(students)         );          renderTable();     }  } |
```

7. Output

The screenshot displays the 'Student Portal Access' login interface. At the top, a dark blue header contains the Symbiosis Institute of Technology logo and name, along with navigation links for Home, Admissions, Academics, and Contact. The main content area is light gray and features a central white login box. This box has a title 'Student Portal Access' and a subtitle 'Enter your Enrollment Number and Password to continue'. It includes two input fields: 'Enrollment Number' (containing '24070521062') and 'Password' (masked with dots). Below the password field is a 'Remember me' checkbox and a 'Forgot Password?' link. A dark blue 'LOGIN' button is positioned at the bottom of the form. A footer note suggests contacting 'helpdesk@sitpune.edu.in' for technical support. The bottom of the page has a dark blue footer with contact information, quick links (Notice Board, Timetable, Examination), and social media links (Facebook, LinkedIn, Twitter), followed by a copyright notice for 2020.

Symbiosis Institute of Technology
Symbiosis International (Deemed University), Pune

Home Admissions Academics Contact

Student Portal Access
Enter your Enrollment Number and Password to continue

Enrollment Number
24070521062

Password
.....

☐ Remember me [Forgot Password?](#)

LOGIN

For technical support, contact helpdesk@sitpune.edu.in

Symbiosis Institute of Technology
Deemed to be University, Pune - 411004
Maharashtra, India

Quick Links
[Notice Board](#)
[Timetable](#)
[Examination](#)

Connect
[Facebook](#)
[LinkedIn](#)
[Twitter](#)

© 2020 Symbiosis Institute of Technology. All Rights Reserved

8. Result / Conclusion

Successfully implemented the grading system and login page with password validation.